| $x_i$ | $d_i = x_i - \bar{x}$ $= x - 15$ | $d_i^2$ |
|---|---|---|
| 11.4 | −3.6 | 12.96 |
| 12.5 | −2.5 | 6.25 |
| 12.8 | −2.2 | 4.84 |
| 16.3 | 1.3 | 1.69 |
| 17.8 | 2.8 | 7.84 |
| 19.2 | 4.2 | 17.64 |
| | | $\Sigma d_i^2 = 51.22$ |

Standard deviation $\sigma = \sqrt{\dfrac{\Sigma d_i^2}{n}}$

$= \sqrt{\dfrac{51.22}{6}} = \sqrt{8.53}$

Hence, $\sigma \simeq 2.9$

**(iii) Assumed Mean method**

When the mean value is not an integer (since calculations are very tedious in decimal form) then it is better to use the assumed mean method to find the standard deviation.

Let $x_1, x_2, x_3, \ldots, x_n$ be the given data values and let $\bar{x}$ be their mean.

Let $d_i$ be the deviation of $x_i$ from the assumed mean $A$, which is usually the middle value or near the middle value of the given data.

$$d_i = x_i - A \text{ gives, } x_i = d_i + A \qquad \ldots(1)$$

$$\Sigma d_i = \Sigma(x_i - A)$$

$$= \Sigma x_i - (A + A + A + \cdots \text{ to } n \text{ times})$$

$$\Sigma d_i = \Sigma x_i - A \times n$$

$$\frac{\Sigma d_i}{n} = \frac{\Sigma x_i}{n} - A$$

$$\bar{d} = \bar{x} - A \text{ (or) } \bar{x} = \bar{d} + A \qquad \ldots(2)$$

Standard deviation $\sigma = \sqrt{\dfrac{\Sigma(x_i - \bar{x})^2}{n}} = \sqrt{\dfrac{\Sigma(d_i + A - \bar{d} - A)^2}{n}}$ (using (1) and (2))

$$= \sqrt{\frac{\Sigma(d_i - \bar{d})^2}{n}} = \sqrt{\frac{\Sigma(d_i^2 - 2d_i \times \bar{d} + \bar{d}^2)}{n}}$$

$$= \sqrt{\frac{\Sigma d_i^2}{n} - 2\bar{d}\,\frac{\Sigma d_i}{n} + \frac{\bar{d}^2}{n}\,(1 + 1 + 1 + \cdots \text{ to } n \text{ times})}$$

$$= \sqrt{\frac{\Sigma d_i^2}{n} - 2\bar{d} \times \bar{d} + \frac{\bar{d}^2}{n} \times n} \qquad \text{(since } \bar{d} \text{ is a constant)}$$

$$= \sqrt{\frac{\Sigma d_i^2}{n} - \bar{d}^2}$$

Standard deviation $\sigma = \sqrt{\dfrac{\Sigma d_i^2}{n} - \left(\dfrac{\Sigma d_i}{n}\right)^2}$

**Thinking Corner**

For any collection of $n$ values, can you find the value of

(i) $\Sigma(x_i - \bar{x})$ (ii) $(\Sigma x_i) - \bar{x}$



10th_Maths_Chapter 8_English.indd 307 12/16/2024 10:06:28 AM